

Global Quantum Mechanics Challenge

Qualification Round 2026

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Problem A

(a) Radio waves ; (b) Infrared ; (c) Ultraviolet ; (d) X-rays ; (e) Gamma rays

Problem B

Given,

- Speed of light $c = 3 \times 10^8$ m/s
- Wavelength of green light $\lambda_g = 532$ nm $= 5.32 \times 10^{-7}$ m
- Wavelength of red light $\lambda_r = 650$ nm $= 6.5 \times 10^{-7}$ m

We know,

- Planck constant $h = 6.626 \times 10^{-34}$ J s
- 1 eV $= 1.602 \times 10^{-19}$ J

Energy of red light photon, $E = \frac{hc}{\lambda_r} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{6.5 \times 10^{-7}}$ J $= 3.058 \times 10^{-19}$ J $= 1.9$ eV

Energy of green light photon, $E = \frac{hc}{\lambda_g} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{5.32 \times 10^{-7}}$ J $= 3.7365 \times 10^{-19}$ J $= 2.3$ eV

The Energy of Red and Green light photon are 1.9 eV and 2.3 eV respectively.

Problem C

(a)

Given,

- Work function $W = 2.3$ eV $= 3.685 \times 10^{-19}$ J

We know,

- Planck constant $h = 6.626 \times 10^{-34}$ J s

If f = the minimum frequency of a photon to make an electron leave from a metal surface, therefore : $W = hf$

$$\therefore f = \frac{W}{h} = \frac{3.685 \times 10^{-19}}{6.626 \times 10^{-34}} \text{ Hz} = 5.56 \times 10^{14} \text{ Hz}$$

The minimum frequency of a photon is 5.56×10^{14} Hz

(b)

Given,

- Work function $W = 2.3$ eV $= 3.685 \times 10^{-19}$ J
- Wavelength of Photon $\lambda = 2500$ Å $= 2.5 \times 10^{-7}$ m

We know,

- Planck constant $h = 6.626 \times 10^{-34}$ J s
- Speed of light $c = 3 \times 10^8$ m/s

Energy of Photon, $E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{2.5 \times 10^{-7}}$ J $= 7.95 \times 10^{-19}$ J $= 4.96$ eV

The maximum kinetic energy $K_{max} = E - W = 4.96 - 2.3$ eV $= 2.66$ eV ≈ 2.7 eV

The maximum kinetic energy of the emitted photoelectron is 2.7 eV

Problem D

(a)

Given,

A photoelectron moving at $v = 5 \times 10^5$ m/s

We know,

Mass of electron $m = 9.11 \times 10^{-31}$ kg

We estimate the uncertainty of velocity $\Delta v \approx v$

Then, the uncertainty of momentum is, $\Delta p = mv$

According to position-momentum uncertainty principle, $\Delta x \Delta p \geq \frac{h}{4\pi}$

Therefore, $\Delta x \geq \frac{h}{4\pi \times mv} = \frac{h}{4\pi \times 9.11 \times 10^{-31} \text{kg} \times 5 \times 10^5 \text{m/s}}$

$\therefore \Delta x \geq 10^{-10}$ m

The estimated uncertainty of position is equal or greater than roughly 10^{-10} m

(b)

Given,

A person is moving at 3 m/s

Mass of person $m = 75$ kg

We estimate the uncertainty of velocity $\Delta v \approx v$

Then, the uncertainty of momentum is, $\Delta p = mv$

According to position-momentum uncertainty principle, $\Delta x \Delta p \geq \frac{h}{4\pi}$

Therefore, $\Delta x \geq \frac{h}{4\pi \times mv} = \frac{h}{4\pi \times 75 \text{kg} \times 3 \text{m/s}}$

$\therefore \Delta x \geq 2.3 \times 10^{-37}$ m

The estimated uncertainty of position is equal or greater than 2.3×10^{-37} m

Problem E

Quantum computer can be used to do very complex works. It can be used for complex calculations, simulations, AI, Data science, crypto, modeling etc. It works faster than normal computers. The two fields where scientists hope quantum computers will be especially useful, are :

1. **Climate Modeling** : Quantum Computer can be used to Simulate atmospheric and oceanic changes to predict the weather. It can track and map precisely.
2. **Nanotechnology** : Using Quantum Computer, we can simulate and design new Nanomaterials. It can also revolutionize Nanomedicine and Nano-electronics.

Among many fields, Climate Modeling and Nanotechnology are two fields where scientists hope quantum computers will be especially useful.

Laptop and Quantum Computer have different mathematical system. They are not for same tasks. Quantum Computer is also very expensive. Furthermore, quantum hardware is completely unsuited for daily works because it cannot naturally process standard data formats like text or images, operates with high error rates, and requires massive cooling systems running near absolute zero.

So, today's laptops are still better than quantum computers for most everyday tasks because of being economical, simple mechanism and suitable work daily works.